Evidential Remote Physiological Measurement via Uncertainty-aware Fusion of Video and RF

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Abstract

Remote physiological measurement enables the capture of vital signals in a non-contact way, which offers significant potential for various applications. Monitoring these signals is achieved through video cameras or radio frequency (RF) sensors, with recent few methods attempting to fuse both sources to leverage complementary patterns for enhanced accuracy. However, these two modalities operate on distinct principles, where video-based methods detect subtle facial color changes from blood volume variations, while RF-based methods capture subtle body vibration due to heartbeats. In practical applications, they may encounter interference at different occasions. Treating these modalities as equally reliable in all situations can lead to suboptimal fusion. To address this issue, we propose an evidential video-RF fusion framework for robust remote physiological signal measurement. We design an uncertainty regression head for each uni-modality, which estimates uncertainty features together with the corresponding physiological signal in each branch. Then an evidential multi-modal fusion module is employed to dynamically fuse the two modalities according to their uncertainty via joint training. Extensive experiments carried on public and self-collected datasets show that the proposed method not only achieves superior fusion performance on easy data collected under well-controlled environment, it also generalizes well to unseen data which represents challenging practical conditions that one or both sensors are disturbed. The code will be released at https://github.com/Walter-White777/Evidential-phys.

CCS Concepts

• **Computing methodologies** → *Biometrics.*

Keywords

rPPG, radar, multimodal fusion, uncertainty, heart rate

ACM Reference Format:
Jieyi Ge, Zhaodong Sun, Wei Peng, Chenhang Ying, Yuwei Chen, Kui Ren, and Xiaobai Li. 2025. Evidential Remote Physiological Measurement via Uncertainty-aware Fusion of Video and RF. In *Proceedings of Proceedings of the 33rd ACM International Conference on Multimedia (MM'25).* ACM, New York, NY, USA, 9 pages. https://doi.org/10.1145/3746027.3754594

1 Introduction

The heartbeat signal is one of the most essential physiological signals, which contains vital information reflecting people's health and emotional status [40]. Traditional cardiac measurement requires contact sensors, e.g., an electrocardiograph or an oximeter, which are inconvenient and constrain applications. Studies show that cardiac activities can be measured contactless via remote sensors, and there are two major approaches, i.e., using a video camera, or a radio frequency (RF) sensor. The new technology would enable numerous applications facilitating e.g., homecare and autonomous driving, while both approaches face limitations.

The video approach exploits subtle facial color variations caused by pulse-induced local blood volume change, referred to as remote photoplethysmography (rPPG) [34]. Early studies explored hand-crafted features from the facial region of interest (ROI) to recover rPPG signals [26, 33, 37]. Later, with the advancement of deep learning different architectures and learning strategies [3, 28, 30, 31, 39, 41] are developed to improve the measurement. However, *the video approach has inherent drawbacks in poor*

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MM'25, Dublin, Ireland
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ACM ISBN 979-8-4007-2035-2/2025/10
https://doi.org/10.1145/3746027.3754594

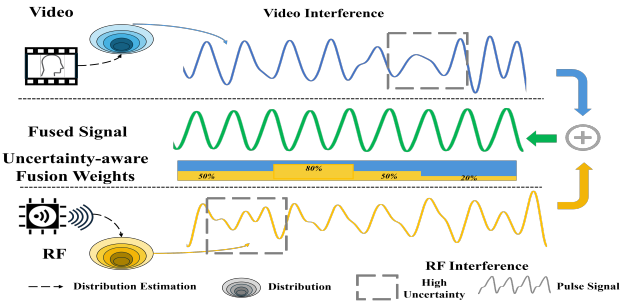


Figure 1: Illustration of the fusion. Both video and RF sensors can be used for remote heartbeat signal measurement. They may encounter disturbance when used alone in practical scenes. We propose uncertainty-aware fusion to dynamically integrate both signals for robust measurement.

illumination conditions, e.g., low ambient light or flashing. An advanced method can hardly reach satisfactory performance itself under such extreme conditions. **The RF approach**, on the other hand, measures cardiac activities by capturing the subtle chest bone vibrations caused by heartbeats. Classical signal processing approaches [22, 24] were used to acquire raw RF signals and followed by filtering techniques to remove irrelevant noise. Subsequently, deep learning-based methods [4, 43] were explored to alleviate interference indivisible by traditional filtering and enhance the quality of heartbeat signals. Although illumination won't impede an RF sensor since it measures motion rather than color, *the RF approach is easily impacted by human motions* since RF purely depends on motion information to measure heartbeat signals.

As shown in fig. 1, the video and RF approach may encounter challenges in practical scenarios when used alone, and it is reasonable to consider integrating the two modalities since they provide complementary information. **The value of video-RF fusion lies in two aspects**: 1) in easy conditions (both modalities are in good quality), the fusion system can exploit complementary information and achieve better performance than each uni-modal; 2) in challenging conditions (significant interferences in one or both modalities), the system can be aware of the input quality and adopt adaptive fusion strategies accordingly to avoid catastrophic performance drop and achieve stable performance.

So far there are two studies [5, 35] worked on video-RF fusion. Vilesov et al. [35] first proposed to integrate the RF modal to address the challenge of dark skin tones in video-based remote physiological measurement. They built and shared the EquiPleth dataset, which contains videos and RF signals from participants with different skin tones. Choi et al. [5] proposed a transformer-based architecture with a temporal cross-attention mechanism for video-RF fusion, but their code and data were not shared. One limitation is that **both studies considered the fusion regardless of the data qualities of each modality.** It is expected that a noisy source input will degrade the final output with such rigid fusion strategies, and the fusion system will easily fail if either the video or RF modality is disturbed.

To address these issues, we propose an evidential video-RF fusion framework for robust remote physiological signal measurement. Specifically, we designed an uncertainty regression head for each uni-modal branch, which estimates uncertainty representation [8, 13] together with the corresponding physiological signal. Then the uni-modal representations are fused dynamically with an Evidential Multi-Modal Fusion (EMMF) module based on their estimated uncertainties. The main contributions of this paper are as follows:

- We propose an evidential multi-modal fusion framework integrating video and RF for remote physiological signal measurement. It is the first work to exploit signal uncertainty for adaptive dynamic fusion of video and RF, which allows robust performance under challenging real-world conditions.
- An uncertainty regression head is designed to model uncertainty in each uni-modal branch, and an evidential multi-modal fusion module is proposed for adaptive fusion of the two modalities.
- We collected a new dataset containing videos and RF signals recorded under various practical human-computer-interactive scenes. Comprehensive experiments are conducted on both public and self-collected datasets. Results indicate that the proposed framework not only achieves superior fusion performance on easy conditions but also achieves robust performance when cross-tested on unseen data with three kinds of challenging conditions.

2 Related Work

2.1 Remote Physiological Measurement

Video-based Approaches Video-based traditional methods explore the subtle color changes on facial regions and extract the periodic signals. Signal decomposition methods like blind source separation [25, 26], and matrix completion [32] were proposed to improve the quality of the recovered rPPG signals. Linear color space transformation [36, 37] estimates the volumetric variations by maximizing the changes induced by diffusion reflection. While these methods have been proved to be efficient for physiological estimation, they largely rely on prior specific knowledge and strong assumptions. As deep learning technology improves by leaps and bounds, spatial-temporal representation of facial videos is fully explored with rPPG tasks. Supervised methods explored different architectures [3, 17, 18, 39, 41] to extract more generalizable representations for estimating accurate pulse signals. Unsupervised methods [10, 28, 30, 31] trained their networks without ground truth pulse signals. Contrastive pairs and rPPG feature distribution regularization were explored and the results have shown to be effective. However, facial videos are restricted by environmental factors like illumination variations and head motions [16], which largely degrade the quality of signals.

RF-based Approaches RF-based methods aim to capture the minus radial depth variations of the chest caused by cardiac activities. Early studies focused on signal decomposition [15, 22, 32] to extract respiration or pulse signals using handcrafted procedures. Later deep learning methods were also proposed for RF-based measurement [4, 11, 43] which lead to advancement in performance. In spite of the improved results, the RF sensors are very sensitive to

electromagnetic interference, e.g., an object in front of the chest could degrade the signal which cannot be fully solved by learning methods.

Video-RF Fusion Vilesov et al. [35] and Choi et al. [5] did preliminary explorations to leverage the complementary properties of the two modalities for remote physiological signal measurement. Viesov et al.[35] collected a large-scale multimodal dataset, the EquiPleth, from subjects with different skin tones. And a three-staged training framework is proposed to predict the plethysmography signals. Choi et al.[5] introduced a dual-branch convolution architecture with a transformer-based fusion strategy to fuse video and RF in temporal domain dynamically. These pioneer studies draw attentions to this valuable yet challenging topic, but a major limitation is that they only consider the fusion in an ideal scene when both inputs are clean and tidy. In practical scenes, one sensor may be interfered with significant signal degradation, which could act as pure noise if directly fused to the other uninfected modality.

2.2 Deep Uncertainty Estimation

Uncertainty obtains growing attention with the increasing application of deep learning techniques, especially in safety-critical scenarios like healthcare or autonomous driving. To model the uncertainty, Bayesian-based methods replace the deterministic parameters of the model with a distribution, as seen in works like [6, 13, 21]. Notably, Kendall and Gal [13] introduced aleatoric uncertainty and epistemic uncertainty to deep learning, addressing data- and model-related uncertainties, respectively. However, the computational cost of Bayesian-based method increases dramatically with enormous parameters on large-scale datasets. To resolve the problem, dropout and deep ensemble methods [9, 14, 38] were utilized to approximate the Bayesian inference. Contrary to the Bayesian neural network, deterministic methods like deep evidential learning [2, 27] efficiently generate uncertainty through a single forward pass. Specifically, Murat et al. [27] placed Dirichlet priors on discrete targets for uncertainty estimation in classification tasks. Then Amini et al. [2] estimated uncertainty in regression tasks with a normal inverse-gamma distribution. With the combination of different distributions, Han et al. [12, 20] extended the single data task to multi-view classification and multi-modal fusion. Inspired by this, we tailored a novel evidential learning method for remote physiological signal measurement task and dynamically fuse the video and RF modalities based on the estimated uncertainty of each modality.

3 Method

As illustrated in fig. 2, the model comprises two branches to facilitate multimodal fusion, expecting to harness complementary information to enhance physiological measurement predictions. Two branches independently extract features from the video and RF modalities, respectively. Rather than directly predicting a deterministic output, we model the distribution of the physiological measurement, which enables modality confidence monitoring. These uni-modal features are then integrated through an Evidential Multi-Modal Fusion (EMMF) module. This uncertainty-aware approach allows for adaptive adjustments in the contribution of each modality during the fusion process, optimizing overall accuracy.

3.1 Uni-modal Representation

Our framework takes a video clip and a radio frequency signal as input and extracts subtle variations caused by periodic cardiopulmonary activities to recover physiological signals y. For each of the two modalities x_m, we build a separate unimodal feature extractor to encode x_m such that later to predict the modality-specific physiological signals y_m.

For video modality V, the model has an input RGB video with the shape of $T \times C \times H \times W$ where T is the number of frames, C is the number of facial image channels and H and W are height and weight, respectively. We take 3D-CNN-based PhysNet [39] as the video extractor to capture the spatial-temporal features for physiological signal estimation.

For RF modality R, the input comprises radio frequency frames with structured chirps reflected from the human body. To capture subtle radial depth variations due to the heartbeat while suppressing environmental noise, we first apply a discrete Fourier transform to convert the RF signal into a range matrix. Since chest vibrations from cardiac activity are minimal and the phase is highly sensitive to background reflections [19, 43], we select the central range bin with the highest power and apply a window of width h to mitigate interference and preserve the key physiological signal. The resulting $h \times T$ matrix is then split into in-phase and quadrature channels and reshaped into vectors of size $(2 \times h) \times T$. Then the vectors are fed into an encoder-decoder architecture comprised of a series of 1D-CNNs [35] to extract periodic physiological features.

3.2 Uncertainty-Aware Fusion

Once uni-modal representations are established, an uncertainty-aware physiological regression head and a bi-modal fusion module are constructed for evidential physiological prediction. Normally, remote physiological signal measurement is treated as a deterministic regression task, learning a mapping from input x_m to predicted physiological signals y_m. This approach typically minimizes the distance between the predicted signal y_m and ground truth y. However, deterministic regression fails to account for underlying noise or uncertainty in real-world scenarios, which is particularly detrimental in multimodal fusion scenarios, as not always that all modalities contribute positively and inappropriate fusion can even cause modality collapse.

To address this limitation, we introduce an adaptive video-RF fusion method utilizing an uncertainty-aware strategy. Rather than adopting a deterministic model for physiological signals, we use Bayesian statistics to model the distribution of predictions, termed Uni-modal Uncertainty Estimation. Building on this, we develop a more robust multimodal fusion approach called Evidential Multi-Modal Fusion (EMMF).

3.2.1 Uni-modal Uncertainty Estimation. Specifically, we consider the physiological signal y is drawn from a Gaussian distribution $\mathcal{N}(\mu, \sigma^2)$, with an unknown mean μ and a variance σ^2. Here, the mean and variance stand for the predicted physiological signal and errors in remote physiological measurement. To estimate the uncertainty of a physiological signal, we apply the Normal Inverse-Gamma distribution [2] to jointly model the mean and variance of the distribution $\mathcal{N}(\mu, \sigma^2)$. As discussed in [2], (μ, σ^2) can be

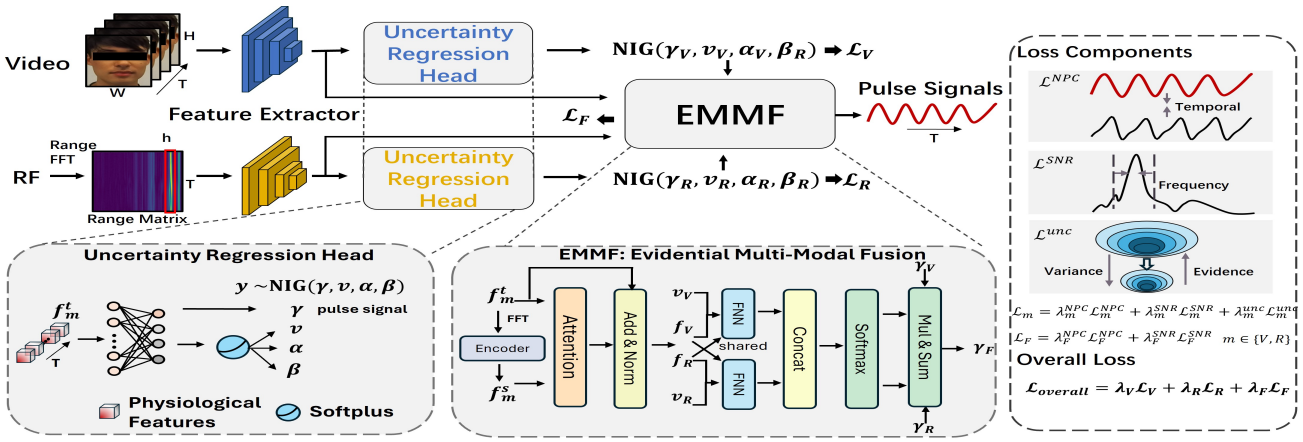


Figure 2: Overview of the proposed framework. The model consists of two branches each first extract physiological features from video or RF inputs. Rather than predicting a single deterministic output, an uncertainty regression head models the distribution of the predictions as NIG(γ, v, α, β), in which γ represents in predicted pulse signal and the rest as uncertainty features. Then an EMMF module adaptively integrates the uni-modal features leveraging uncertainty to achieve the fused pulse signal γ_F. FNN represents feed-forward neural network and two FNNs share parameters. Three loss components regulate the model on both uni-modal and fusion levels via joint training. V, R, and F indicate video, RF, and fusion, respectively. NIG, NPC, SNR, and Unc indicate normal inverse gamma, negative Pearson correlation, signal noise ratio, and uncertainty, respectively. Subscript m represents uni-modality.

modeled with a Gaussian prior and an Inverse-Gamma prior as follows,

$$y \sim \mathcal{N}(\mu, \sigma^2), \tag{1}$$

$$\mu \sim \mathcal{N}(\gamma, \sigma^2 v^{-1}), \sigma^2 \sim \Gamma^{-1}(\alpha, \beta), \tag{2}$$

where $\Gamma(\cdot)$ denotes the gamma function, $\gamma \in \mathbb{R}, \sigma > 0, \alpha > 1, \beta > 0$.

Therefore, the goal is to estimate the mean μ and the variance σ^2, which can be formulated as a posterior distribution $q(\mu, \sigma^2) = p(\mu, \sigma^2|y)$. Based on the assumption that the mean and the variance are independent, $q(\mu, \sigma^2)$ can be separated into two independent distributions. This allows $q(\mu, \sigma^2)$ can be formulated as a normal inverse-gamma distribution NIG(γ, v, α, β), which is defined as:

$$q(\mu, \sigma^2|\gamma, v, \alpha, \beta) = \frac{\beta^\alpha \sqrt{v}}{\Gamma(\alpha)\sqrt{2\pi\sigma^2}} \left(\frac{1}{\sigma^2}\right)^{\alpha+1} \exp\left\{-\frac{2\beta + v(\gamma-\mu)^2}{2\sigma^2}\right\}. \tag{3}$$

Using this formulation, we design an uncertainty-aware regression head for remote physiological measurement that outputs both pulse signals and the corresponding NIG parameters. The uncertainty regression head consists of a projector that projects physiological features from signal space to parameters space, which outputs $S_{out} \in \mathbb{R}^{T\times 4}$, where T is the length of physiological signals and 4 represents the number of parameters in NIG distribution. S_{out} is split into four parts and then we enforce the constraints on three of them (v, α, γ) with a Softplus activation and finally generate the NIG parameters $(\gamma, v, \alpha, \beta)$. Having established the distribution of evidential learning, we can define the predicted signal with the parameters above:

$$\underbrace{\mathbb{E}[\mu] = \gamma}_{\textbf{predicted signal}} \tag{4}$$

As for parameters estimation, given the target physiological signal y, the model evidence i.e., the likelihood of y is supposed to be maximized, which can be computed by marginalizing over the likelihood parameters. Thus in the training stage, we take the negative logarithm of the model evidence as one of the objectives. According to Bayesian probability theory, the negative logarithm loss (nll) function can be formulated as:

$$\mathcal{L}^{nll}(\boldsymbol{w}) = \frac{1}{2}\log\left(\frac{\pi}{v}\right) - \alpha\log(\Omega) + \left(\alpha + \frac{1}{2}\right)\log((y_i - \gamma)^2 v + \Omega) + \log\left(\frac{\Gamma(\alpha)}{\Gamma(\alpha + \frac{1}{2})}\right), \tag{5}$$

where $\Omega = 2\beta(1+\gamma)$ and $\boldsymbol{w} = (\gamma, v, \alpha, \beta)$. This loss function $\mathcal{L}^{nll}$ directs the model to output the parameters of an NIG distribution that best fits the ground-truth physiological signals, optimizing the model by maximizing evidence.

Besides adapting video or RF data to the ground-truth, the existence of noisy data such as facial videos with varying light or RF signals interfered with occlusions will largely influence the performance of remote physiological measurement. Therefore evidence of inaccurate predictions from these noisy data also needs to be calculated and reduced. To address this, a regularization term is formulated to reduce the evidence where the physiological signal prediction is inaccurate:

$$\mathcal{L}^r(\boldsymbol{w}) = |y - \mathbb{E}(\mu)| \cdot \Phi = |y - \gamma| \cdot (2v + \alpha), \tag{6}$$

where y is the ground truth physiological signal, Φ is the total evidence to measure the confidence of predictions defined by [2]. This loss penalizes the wrong predictions as well as scales the evidence with this error case.

Therefore, the total uncertainty loss for video modality can be defined as follows, and radar modality is the same:

$$\mathcal{L}_V^{unc}(\boldsymbol{w}) = \mathcal{L}_V^{nll}(\boldsymbol{w}) + \tau \mathcal{L}_V^{r}(\boldsymbol{w}), \tag{7}$$

where τ is the coefficient trades off uncertainty inflation with model fitting.

3.2.2 Evidential Multi-Modal Fusion. After obtaining the uni-modal representations for both modalities, we propose a dynamic fusion mechanism based on the uncertainty estimation of RGB video and RF signals. To better model cross-modal uncertainty, our approach not only leverages the confidence scores of individual modalities but also incorporates feature-level interactions, thereby enhancing the reliability and performance of the fused representation.

As illustrated in Fig 2, we first extract spectral features f_m^s from the temporal-domain physiological features f_m^t using Fast Fourier Transform (FFT) followed by a 1D convolutional encoder. This process emphasizes robust spectral components associated with heart rate estimation:

$$f_m^s = \text{Encoder}(\text{PSD}(f_m^t)), \tag{8}$$

where PSD denotes the power spectral density derived from FFT and m represents uni-modality.

The temporal and spectral features are then fused through an attention module to aggregate complementary physiological information. Specifically, both features are projected into a shared latent space, enabling cross-domain interaction:

$$\begin{aligned} &\hat{f_m^t} = \text{Proj}(f_m^t), \quad \hat{f_m^s} = \text{Proj}(f_m^s); \\ &f_m = \text{Norm}\left(f_m^t + \text{Softmax}(\hat{f_m^t}\hat{f_m^s}^T) f_m^s\right), \end{aligned} \tag{9}$$

where Proj denotes a linear projection. This attention-driven fusion facilitates contextual perception of signal characteristics, effectively combining temporal dynamics and spectral patterns to enhance physiological feature representation.

To further exploit cross-modal dependencies, we introduce a learnable fusion mechanism that dynamically adjusts modality contributions based on both intra-modal confidence and inter-modal relationships. The fusion results γ_F is computed as:

$$\gamma_F = \sum_{m \in \{V,R\}} \text{Softmax}\left(\mathbf{v}^T \text{Relu}(\mathbf{W}[f_m; v_m; f_{\neg m}])\right) \gamma_m, \tag{10}$$

where $\mathbf{W}$ and $\mathbf{v}$ are trainable parameters, $f_{\neg m}$ represents features from the complementary modality, and v_m denotes the confidence score of modality m.

By explicitly modeling cross-modal interactions, our fusion mechanism adaptively balances modality-specific uncertainties while capturing latent dependencies between RGB and RF signals. This not only improves robustness against noisy or missing data but also enhances generalization in complex real-world scenarios (e.g., varying illumination or motion artifacts). Moreover, the uncertainty-aware weighting provides interpretable insights into modality reliability, aligning with the principles of trustworthy multimodal learning.

3.3 Dynamic Joint Loss

Besides uncertainty learning, we also add the pulse signals supervision to the temporal and frequency domain of predicted signal γ in eq. (4). Specifically, negative Pearson correlation(NPC) loss [39], $\mathcal{L}^{NPC}$ is adopted to constrain the overall trend of signals, which is easy to converge and overfit on the validation set. In addition, signal-to-noise ratio (SNR) loss [29], $\mathcal{L}^{SNR}$, is further applied to radio frequency to capture the periodicity within the target frequency band, which is hard to converge due to the sensitive characteristic and irrelevant environmental noise, especially for radio frequency modality. To both preserve the overall signal trend and periodic features, we adopt dynamic joint loss as our training strategy to learn from the easier part (NPC) to the harder part (SNR), in which we increase the weight of frequency term gradually with exponential increment strategy. The loss function of video, RF and the fused distribution can be formulated as follows:

$$\begin{aligned} \mathcal{L}_m &= \lambda_m^{NPC}\mathcal{L}_m^{NPC} + \lambda_m^{SNR}\mathcal{L}_m^{SNR} + \lambda_m^{unc}\mathcal{L}_m^{unc}, \\ \mathcal{L}_F &= \lambda_F^{NPC}\mathcal{L}_F^{NPC} + \lambda_F^{SNR}\mathcal{L}_F^{SNR} \end{aligned} \tag{11}$$

where $m \in \{V, R\}$ and $\lambda_m^{NPC}, \lambda_m^{SNR}, \lambda_m^{unc}$ represent the weight of NPC, SNR, uncertainty loss of video and RF, respectively. The details of the weights can be seen in the supplementary materials.

Inspired by multi-task learning, the overall loss can be defined as the sum of unimodal losses $\mathcal{L}_R$, $\mathcal{L}_V$, and fusion loss $\mathcal{L}_F$:

$$\mathcal{L} = \lambda_R \mathcal{L}_R + \lambda_V \mathcal{L}_V + \lambda_F \mathcal{L}_F \tag{12}$$

where $\lambda_R = 0.1, \lambda_V = \lambda_F = 1$.

4 Experimental Results

4.1 Datasets and Experimental Settings

EquiPleth [35] is the only public RGB-RF dataset for remote physiological signal measurement with three different skin tones (light, medium, and dark). It contains 550 recordings of 30-second videos and radar data from 91 subjects and six recordings were taken for each subject. It was collected using a Zeda stereo camera and a Texas Instruments AWR1443 RF development board at a distance of 0.5 to one meter. The ground truth plethysmography signals were acquired using an IntelliVue MX800 clinical-grade patient monitor. EquiPleth concerns different populations but lacks environmental variations, and the data are simple since the subjects were still with no tasks. We collected a new dataset involves practical application scenarios.

MPU-Phys is a newly collected multi-modality dataset for video-RF remote physiological measurement which contains two parts. **MPU-Phys-A** represents ordinary human-computer-interactive (HCI) scenes, when subjects watch movies from a screen with indoor lighting. Videos and RF data of 22 subjects were recorded simultaneously for about 24 minutes. The RGB videos and RF signals were captured with a Logitech webcam and Texas Instruments AWR1843 development board, respectively. Ground truth photoplethysmograph signals were recorded using a MindRay ePM monitor. **MPU-Phys-B** represents three challenging scenarios that may encounter in practical HCI scenes, i.e., *1) varying light:* flashing reflects on the screen in a dark room (impact video but not RF); *2) RF Occlusion:* an iPad placed in front of the chest (impact RF but not video);*3) motion:* playing a game using the keyboard (may impact both RF and video) Ten subjects were recorded for ten minutes, i.e., two minutes for each of the three scenarios with two-minute normal condition in between. The same devices were used as part

Table 1: Intra-dataset results on EquiPleth and the MPU-Phys-A. The best results are marked in bold and the second best results are underlined.

Method	Modality	EquiPleth			MPU-Phys-A		
		MAE↓	RMSE↓	r↑	MAE↓	RMSE↓	r↑
CHROM [7]	Video	7.31	17.42	0.54	4.49	8.30	0.60
POS [37]	Video	4.64	15.07	0.68	3.55	6.68	0.73
PhysFormer [41]	Video	1.47	4.76	0.92	4.44	6.35	0.64
Contrast-Phys+(100%) [31]	Video	1.16	5.71	0.87	2.81	4.44	0.84
Vilesov et al. [35]	Video	1.78	5.26	0.91	3.08	5.46	0.79
Ours(RGB-only)	Video	1.39	4.32	0.93	**2.60**	4.42	0.81
FFT-based [1]	RF	13.26	21.00	0.23	14.69	20.47	0.02
Vilesov et al. [35]	RF	2.18	6.12	0.89	4.51	7.58	0.65
Ours(RF-only)	RF	2.02	5.83	0.89	4.15	6.67	0.70
Vilesov et al. [35]	Fusion	1.12	3.42	0.95	4.71	6.59	0.76
Ours	Fusion	**0.79**	**2.37**	**0.98**	2.64	**4.00**	**0.88**

A.

Implementation Details: The proposed method is implemented with Pytorch and trained on a Nvidia H100 GPU in an end-to-end manner. The model is trained for 40 epochs with AdamW optimizer and learning rate and weight decay are 1e-4 and 1e-6, respectively. The inputs of the model are segmented into 10-second clips during training phase. For testing phase, we follow Vilesov's protocol [35],adopting a 10s clip for EquiPleth. As for MPU-Phys, we follow [31], that 30s clip for testing. Specifically, we use the MTCNN face detector [42] to crop the centered face area and resize to 128×128 for video branch. For RF branch, the window size of the maximum power occupancy of the range matrix $h = 5$. The regularization weight in the uncertainty loss $\tau = 0.5$.

Tasks and Evaluation Metrics: Three experiments were conducted. First, intra-dataset heart rate estimation was tested on EquiPleth [35] and MPU-Phys-A datasets in 4.2. Second, we conduct ablation study on EquiPleth dataset to verify the effectiveness of each designed modules in 4.3. Third, the method is further tested on MPU-Phy-B dataset representing challenging recording conditions, and in a cross-dataset test mode (trained on EquiPleth) to demonstrate its generalization ability. Uncertainty variations are visualized to verify the validity of the evidential fusion strategy. For the protocols, we follow the same way as in [35] on EquiPleth dataset. For MPU-Phys-A, we divide it into training and testing set with a ratio of 7:3. As for MPU-Phys-B, we employ a cross-validation approach where half the dataset is used to fine-tune the model pretrained on EquiPleth, while the remaining half serves as the test set. For evaluation metrics, we follow previous works [16, 23] and take the mean absolute error (MAE), root mean squared error (RMSE) and Pearson correlation coefficient r to evaluate the performance of physiological signal measurement.

4.2 Uni-modal and Fusion Results

We test the proposed framework on EquiPleth and MPU-Phy-A datasets in an intra-dataset evaluation mode, i.e., the model was trained and tested on the same dataset with a pre-defined protocol. Results of uni-modals and fusion are listed in Table 1. We compare the performance with previous methods on both uni-modality and fusion modality. It can be observed that on both datasets, our method achieves consistent improvements comparing to the baseline approaches across all three evaluation metrics under uni-modal and fusion settings, which demonstrates the superiority of the proposed method. The results underscore the effectiveness of predicted normal inverse-gamma distribution as it naturally fits to regression tasks, whereas previous works solely predict the physiological signals. The objectives in uni-modal uncertainty estimation also facilitate the model to fit the target signals from the distribution perspective and reduce the evidence on the wrong predictions, thus output a more stable mean of the predicted pulse signals.

Comparing the performance of our method among different modalities, it is observed that video-RF fusion consistently outperforms each single modality on both dataset, which verifies the effectiveness of the proposed fusion strategy from the first aspect (see Section 1), i.e., under easy conditions when both modalities presents comparatively good quality inputs, the proposed evidential fusion strategy can better exploit the two modalities and achieve superior performance than using single modality. While the fusion advantage is not guaranteed for all fusions, as for [35] there are obvious fusion performance drops on the MPU-Phys-A dataset. While the MAE of our method is comparable to that of SOTA methods, we observe significant improvements in RMSE and r, demonstrating its advantages.

4.3 Ablation Study

Effect of the Uncertainty Regression Head and EMMF

The uncertainty regression head (for uni-modality) and the EMMF are the two core modules in our framework which are evaluated separately and the quantitative results are reported in Table 2.

For each ablation setting, we train the model under identical conditions for the same modality to ensure that the observed performance changes are due to the presence or absence of the specific components. The two backbones here indicate the processes used in Vilesov et al [35]. Before the fusion module is involved, every

Table 2: Ablation of the Uncertainty Regression Head and EMMF on EquiPleth. ● and ▲ represent video and RF modality, respectively. The last row shows the results after fusion and the best results are marked in bold.

Backbone	Uncertainty Regression Head	EMMF	MAE↓	RMSE↓	r↑
✓			●1.78	5.26	0.91
			▲2.18	6.12	0.89
✓	✓		● 1.39	4.32	0.93
			▲2.02	5.83	0.89
✓	✓	✓	**0.79**	**2.37**	**0.98**

uni-modal branch generates results separately as listed in the first four rows.

From the results shown in the first four rows, it can be seen that the performance are improved on both video and RF modalities across all three metrics after the uncertainty regression head is applied, which demonstrated the effectiveness of the designed module.

From the last row of results it can be observed that there is a significant performance elevation after the EMMF module is involved, that the errors have been reduced by about 35% in terms of the RMSE and the MAE compared to the best performing uni-modality (video). The results show the EMMF module can effectively leverage complementary information from the two modalities for dynamic fusion, thereby achieving superior performance.

Fusion Strategy Comparison

To further verify the effectiveness of the proposed evidential fusion strategy, we compare it to two other fusion methods adopted in previous video-RF fusion work, i.e., additive fusion used in [35], and cross-attention fusion used in [5]. The comparison was conducted again on EquiPleth dataset.

Table 3: Comparison of different fusion strategies.

Fusion Method	MAE↓	RMSE↓	r↑
Additive Fusion [35]	1.84	7.12	0.82
Cross-Attention [5]	1.06	3.68	0.94
Evidential Fusion(ours)	**0.79**	**2.37**	**0.98**

Results in Table 3 show the superior performance of the evidential fusion over the other two fusion strategies. To be specific, the additive fusion is a late-fusion strategy which integrates the two modalities by averaging the pulse signals from each uni-modality. This vanilla fusion method cannot achieve satisfactory results and even performs worse than the single video modality. As for cross-attention based fusion, we compute attention scores between the two modal sequences in the latent space to obtain their semantic relations which are used to achieve the final fused signal. Cross-attention performs better than the additive fusion and the errors are smaller than those of uni-modality results, but the mechanism is hard to interpret and the confidence of these two modalities in different scenarios is ignored. The proposed evidential fusion explicitly outperform the other two on all three metrics, which attributes to the full exploit of data uncertainty and dynamic fusion of these two modalities.

4.4 Evaluation under Challenging conditions

We conduct cross-dataset experiment to further test the generalization ability and robustness of the proposed method. The model is trained on EquiPleth dataset and tested on MPU-Phys-B dataset representing three challenging scenarios when one or both modalities are degraded due to interferences.

We first visualize the fusion weight and predicted pulse waveforms of the video and RF across different scenarios to verify that the designed scenarios did interfered target modalities as expected. In addition, it is also designed to validate our method is aware of the interference and dynamically adjust the weight of two modalities during fusion.

We select 30 seconds clip of different scenarios and visualize the weights of two modalities in Fig. 3. In normal environments, both modalities operate without significant interference, leading to a dynamically adjusted fusion strategy that balances the contributions of video and radar signals. The model leverages the complementary nature of both modalities to optimize performance. Under occlusion conditions(ipad in front of the chest), the radar modality is predominantly affected, resulting in an increased reliance on the video signal. This adaptive adjustment assigns a higher fusion weight to the video, compensating for the degradation in radar signal quality. In scenarios with varying lighting, the video modality is primarily impacted, prompting the model to assign a higher fusion weight to the radar signal when the light is changing. Since the radar does not depend on lighting, the radar maintains relative stability, providing a more accurate representation of physiological signals despite lighting changes. During motion scenarios, both modalities are influenced by head and body movements, leading to a more uniform distribution of fusion weights. This approach ensures that the model can effectively manage the reduced reliability of both signals, thereby maintaining overall performance.

Fig. 4 displays the pulse waveforms corresponding to these challenging scenarios. The first two rows demonstrate a clear correlation between fusion weights and rPPG signal quality, reinforcing that the fusion mechanism dynamically adjusts based on modality reliability. The last row shows the evidential fusion can reconstruct relatively stable outputs.

Then we report quantitative results in Table 4. MAE is adopted as the metric to evaluate the measurement performance. From the first row, it can be seen that under all three challenging conditions the vanilla fusion suffers large performance drop with MAE values over 10 or even 20, indicating its vulnerability under such practical conditions as expected. In contrast, evidential fusion results in the last row indicate that our method can generalize well to unseen data, and it can maintain satisfactory performance even when one modality is heavily disrupted as the evidential fusion strategy allows adaptive fusion according to data uncertainty.

More detailed observations can be drawn from the three rows of our results. First, the uni-modal performance curves exhibit patterns closely aligned with the weight variations shown in Fig.3, confirming a direct correlation between estimation errors and fusion

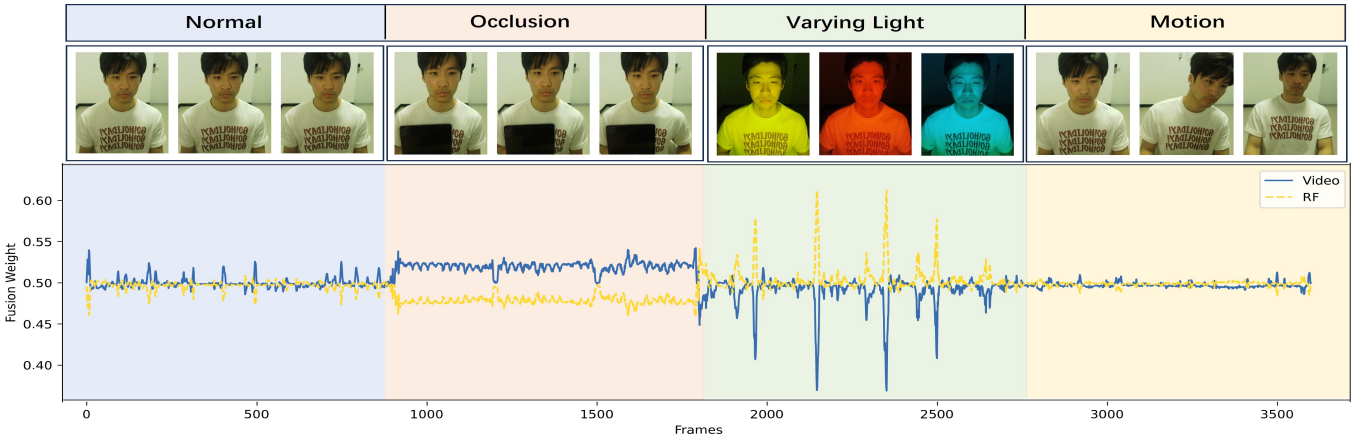


Figure 3: Visualization of fusion weights of Video and RF under different scenarios. The normal scenario serves as the reference without any interference. In the occlusion scenario, an iPad is placed in front of the chest to disrupt RF signals. Under varying light, a strobe light is used to interfere with facial video capture. In the motion scenario, head and body movements introduce interference to both video and radar RF signals. The curves shows our method dynamically adjust the fusion weight of two modalities.

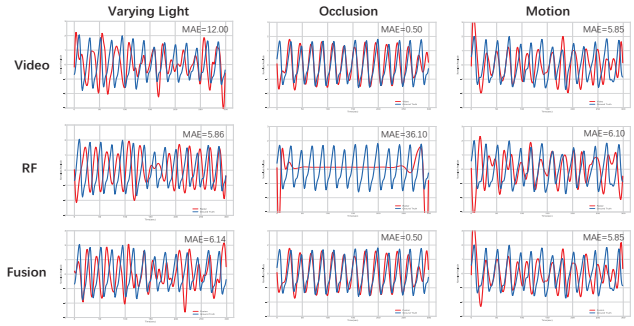


Figure 4: Pulse Waveforms visualization in different scenarios: varying light, occlusion and motion, respectively. The ground truth signals are represented in blue and the predicted signals are represented in red.

Table 4: Test results on the challenging MPU-Phys-B dataset. We use half to finetune the model pretrained on EquiPleth and the rest for test in a cross-validation manner. Results show that our method gets the best results under most of the challenging environments, which are much better than the previous vanilla fusion method [35].

Method	MPU-Phys-B (MAE↓)			
	Varying Light	Occlusion	Motion	Normal
Vilesov et al. [35]	10.29	25.71	23.14	7.71
RF	**2.98**	13.38	7.76	2.69
Video	7.68	2.86	7.48	2.63
Evidential Fusion	3.40	**2.51**	**7.27**	**2.40**

weights. This suggests that the fusion mechanism effectively adapts to the reliability of each modality. Second, evidential fusion demonstrates robust performance across most challenging scenarios, with the exception of motion interference. This limitation is reasonable because motion artifacts degrade both modalities throughout the recording, making it difficult to extract reliable information when both signals are compromised. There is a tiny margin of the RF performance over fusion under the varying light condition. It might be caused by over-reliance on the video modality during training with the EquiPleth dataset [35], on which video modality always yields better performance as shown in previous results tables.

5 Conclusions

In this paper, we propose an uncertainty-aware fusion framework for remote physiological measurement based on RGB videos and RF signals. With the help of Normal Inverse-Gamma distribution, we model the physiological signal regression with four parameters and design an uncertainty regression head for each uni-modality to estimate these parameters. Followed with an Evidential Multi-Modal Fusion (EMMF) module, our method can dynamically fuse these modalities based on their confidence. Extensive experiments were conducted on the public EquiPleth dataset and a newly constructed MPU-Phys dataset representing ordinary HCI scenes and challenging recording conditions when one or both modalities are significantly disturbed. Results demonstrate that our method can effectively capture the uncertainty in physiological signals and provide evidence for robust fusion in challenging scenarios.

Acknowledgments

This work was supported by the Startup Foundation for Introducing Talent of NUIST and the Research Funds of Hangzhou Institute of Advanced Study, UCAS (B02006C021026).

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